

Basic Integration by Substitution*

Benjamin Abshire
Kent State University

February 19, 2024

1 Definition

If $u = g(x)$ is a differentiable function whose range is an interval I and f is continuous on I , then

$$\int f(g(x)) g'(x) dx = \int f(u) du$$

Putting it another way:

If $F'(x) = f(x)$ and $g(x)$ is a function, then $\frac{d}{dx} F(g(x)) = F'(g(x)) g'(x) = f(g(x)) g'(x)$. Where F is an anti-derivative of f . So,

$$\int f(g(x)) g'(x) dx = F(g(x)) + C$$

Now all together:

$$\int f(g(x)) g'(x) dx = \int f(u) du = F(u) + C = F(g(x)) + C$$

We want to set u equal to a function whose derivative is also present in the integral. u will always be the “inside” function in our composite function. du is the derivative of u . It will always contain dx and any other term not included in u . Between u and du , every term in the integral should be accounted for.

*Definitions and Examples from Dr. Donald White's Lecture and Stewart's *Essential Calculus* Second Edition

2 Example

$$\int 2x (x^2 + 4)^7 dx$$

Our “inside function” here is $x^2 + 4$ so we will set it equal to u . The derivative of u is equal to $2x dx$:

$$\text{Let } u = x^2 + 4, du = 2x dx$$

$$\text{So, } \int 2x (x^2 + 4)^7 dx = \int (u)^7 du$$

Next, we take the anti-derivative of $(u)^7$:

$$\frac{1}{8}u^8 + C$$

Subbing back in for u :

$$\frac{1}{8}(x^2 + 4)^8 + C$$